



Naked-eye view

Dark and light features are visible. The full Moon looks like a face, with Mare Imbrium and Mare Serenitatis as eyes, and Mare Nubium and Mare Cognitum as the mouth.



Binocular view

The Moon is still seen as a whole, but a well-supported pair of binoculars reveal surface features, such as large craters. The uneven nature of the terminator (see opposite) is also visible.



Telescope view

Now only a part of the Moon is visible. Thousands of smaller craters and details of shadows, mountains, and valleys become distinct. The usable magnification depends on the turbulence of Earth's atmosphere.

Observing the Moon

The Moon's surface is dry, dusty, dark, and dead. The lunar landscape has remained virtually unchanged for millions of years. A "new" crater, wider than half a mile (1 km) across, is formed on the Earth-facing side on average every 40,000 years. A glance at the Moon will reveal that it has two types of terrain: large dark plains, called maria (mare in the singular); and brighter, heavily cratered highland regions.

THE NEAR SIDE

Very early in the Moon's life, when it was much closer to Earth and the interior was hotter, it became locked so that one face—the near side—always pointed toward Earth. This affected the crust, and not only is the near side, on average, 3 miles (5 km) lower than the far side, the near-side crust is also about 15 miles (25 km) thinner. Deep craters on the near side were filled with

volcanic lava in their early history; this did not happen on the far side. Dark lava plains now cover about half the near side. When the Moon was first mapped, astronomers thought these dark regions were water and referred to each as a mare (sea) or an oceanus (ocean). It is useful to learn their names and positions.

More recently formed craters can be seen within the dark, flat maria. The higher mountainous regions, which are older and have more craters, are twice as bright as the maria.



Near-side view

The near side is dominated by the large, dark Oceanus Procellarum on the western side. Bright spots in the region are more recently formed craters. Copernicus is particularly prominent.